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RYO KAMOI

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https://scholar.google.com/citations?user=4OWTLKAAAAAJ

Final-year PhD student at Penn State University, supported by the Google PhD Fellowship. My research interests lie in:

- Self-Evolution of LLMs and LLM Agents, Agent Harness Improvement, and Post-Training [TACL'24, COLM'24]
- LLM-Based Verification of LLM Responses and Intermediate Steps, Formal Verification [ACL'26 Findings]
- LLM-as-a-Judge Evaluation of Open-Ended and Non-Verifiable Tasks, Including Deep Research, Simulated Conversations [EMNLP'26], and Faithfulness Evaluation [EMNLP'23, EACL'23]

Education

Pennsylvania State University - Ph.D. Candidate in Computer Science — Aug 2023 - May 2027 (expected)
Advisor: Dr. Rui Zhang

University of Texas at Austin - Master of Science in Computer Science — Aug 2020 - Dec 2022
Advisor: Dr. Greg Durrett, Mentor: Dr. Tanya Goyal

Keio University, Japan - Bachelor of Engineering in Statistics — Apr 2016 - Mar 2020
Advisor: Dr. Kei Kobayashi

Research Internships

Google, Sunnyvale, CA – Student Researcher at the Cloud AI Research team — Jun - Sep 2026
Research on agent harness improvements for LLM agents that improve rubric-based LLM judges

Microsoft, Redmond, WA – Research Intern at the Office of Applied Research — May - Dec 2025
Research on LLM-based conversation simulation for chatbot training environments

Amazon, Cambridge, UK – Applied Scientist Intern at the Alexa team — Jul - Dec 2021
Research on detecting errors in LLM chatbot responses

Selected Papers

https://scholar.google.com/citations?user=4OWTLKAAAAAJ

Ryo Kamoi, Yanfei Chen, Rujun Han, Yufan Zhuang, Rui Zhang, Chen-Yu Lee, Tomas Pfister. RubricRefiner: A Self-Evolving Agent for Improving Rubric-Based LLM Judges. 2026.

Ryo Kamoi, Ameya Godbole, Binglin Zhou, Xiaoxin Lu, Longqi Yang, Rui Zhang, Mengting Wan, Pei Zhou. 2026. Evaluating LLM-Simulated Conversations in Modeling Inconsistent and Uncollaborative Behaviors in Human Social Interaction. *EMNLP 2026 (main)*.

Ryo Kamoi, Yusen Zhang, Nan Zhang, Sarkar Snigdha Sarathi Das, Ranran Haoran Zhang, Wenpeng Yin, Rui Zhang. 2026. Efficient PRM Training Data Synthesis via Formal Verification. *ACL 2026 Findings*.

Ryo Kamoi, Yusen Zhang, Sarkar Snigdha Sarathi Das, Ranran Haoran Zhang, Rui Zhang. 2025. VisOnlyQA: Large Vision Language Models Still Struggle with Visual Perception of Geometric Information. *COLM 2025*.

Ryo Kamoi, Yusen Zhang, Nan Zhang, Jiawei Han, Rui Zhang. 2024. When Can LLMs Actually Correct Their Own Mistakes? A Critical Survey of Self-Correction of LLMs. *TACL 2024*.

Ryo Kamoi, Sarkar Snigdha Sarathi Das, Renze Lou, Jihyun Janice Ahn, Yilun Zhao, Xiaoxin Lu, Nan Zhang, Yusen Zhang, Ranran Haoran Zhang, Sujeeth Reddy Vummanthala, Salika Dave, Shaobo Qin, Arman Cohan, Wenpeng Yin, Rui Zhang. 2024. Evaluating LLMs at Detecting Errors in LLM Responses. *COLM 2024*.

Ryo Kamoi, Tanya Goyal, Juan Diego Rodriguez, and Greg Durrett. 2023. WiCE: Real-world Entailment for Claims in Wikipedia. *EMNLP 2023 (main)*.

Honors and Awards

Scholarships & Academic Honors

- 2026 Google North America PhD Fellowship in Natural Language Processing — 2026
- 2026-27 Herbert J. Bailey Graduate Fellowship, Penn State University — 2026
- 2025-26 Vice Provost and Dean of the Graduate School Student Persistence Scholarship, Penn State University — 2026
- Keio University Global Fellowship - Scholarships for my graduate study at UT Austin — 2020
- Keio Engineering Foundation Award - Graduation with highest honors (First place in the Dept. of Mathematics) — 2020
- Japan Student Services Organization (JASSO) Exchange Student Scholarship — 2018

Research Papers

- First Place in the Poster Session at industryXchange 2026 — 2026
- Best Paper Award at the 2nd AI4Research workshop @ AAAI 2025 — 2025

Services

- Great Reviewer at ACL 2026 (133/8,594) — 2026
- Top 200 Reviewer at ICLR 2026 (200/18,054) — 2026
- Outstanding Reviewer at EMNLP 2024 — 2024

Media Mentions

Self-Correction in Large Language Models. **Communications of the ACM**. Feb 26, 2025

Academic Services and Educational Experience

Reviewer (*outstanding reviewer)

ARR (2023-present), COLM 2024-present, **ACL 2026***, **ICLR 2026***, NeurIPS 2025, ICCV 2025, CVPR 2025, **EMNLP 2024***, AISTATS 2021, BMVC 2020

Conferences and Workshops - MASC-SLL 2025 (organizer), EMNLP 2024 (volunteer)

Events - NLP Colloquium JP (organizer, 2024 -)

Teaching Assistant, Penn State University

- CMPSC 442: Artificial Intelligence — Spring 2024
- CMPSC 448: Machine Learning and AI — Fall 2023, Fall 2024, Fall 2025

Invited Talks

Towards Self-Evolving LLMs

- AiTech Seminar at Okazaki Laboratory, Institute of Science Tokyo — 2025
- Michinoku Communication Science Seminar (MiCS) at Tohoku University — 2025
- AI for Math Minisymposium at MSIAM-NNP — 2025

Emergence and Limitations in Self-Refinement of LLMs

- Matuo-Iwasawa Lab at the University of Tokyo — 2024

Automatic Detection of Mistakes in LLM Responses

- NLP Colloquium JP — 2023
- Nagoya NLP Seminar at Nagoya University — 2023

Other Papers

First-Author Papers

Ryo Kamoi, Tanya Goyal, Juan Diego Rodriguez, and Greg Durrett. 2023. Shortcomings of Question Answering Based Factuality Frameworks for Error Localization. *EACL 2023 (main)*.

Ryo Kamoi and Kaname Tomite. 2021. Efficient Unknown Object Detection with Discrepancy Networks for Semantic Segmentation. *In the NeurIPS 2021 Workshop on Machine Learning for Autonomous Driving.*

Ryo Kamoi and Kei Kobayashi. 2020. Why is the Mahalanobis Distance Effective for Anomaly Detection? *arXiv preprint arXiv:2003.00402*.

Ryo Kamoi and Kei Kobayashi. 2020. Out-of-Distribution Detection with Likelihoods Assigned by Deep Generative Models Using Multimodal Prior Distributions. *In The AAAI's Workshop on Artificial Intelligence Safety*.

Ryo Kamoi, Kei Kobayashi. 2019. Likelihood Assignment for Out-of-Distribution Inputs in Deep Generative Models is Sensitive to Prior Distribution Choice. *arXiv preprint arXiv:1911.06515.*

Co-Authored Papers

Binglin Zhou, Peng Shi, **Ryo Kamoi**, Nan Zhang, Rui Zhang. 2026. ToolSciVer: Multimodal Scientific Claim Verification with Visual Tool Augmented Reinforcement Learning. arXiv preprint arXiv:2607.16131.

Jihyun Janice Ahn, **Ryo Kamoi**, Berk Atıl, Renze Lou, Wonwoo Kang, Heehyun Park, Sarkar Snigdha Sarathi Das, Zhuoyang Zou, Xiaoxin Lu, Yusen Zhang, Asfahan Shah, Ridwanul Hasan Tanvir, Lingxiao Zhao, Hongxi Huang, Vignesh Venkatesh, Dianjun Lin, Hamid Shah, Wentao Wang, Zhanpeng Song, Joshua Reed Bassin, Dax Patel, Ishan Appareddy Agrahar, Sahil Pardasani, Xin Dong, Fatemeh Rahbari, Benjamin David Rishel, Soochan Andrew Lee, Yuv Boghani, Ali B. AlNaseeb, Pranav Suby, Seokhyeon Bae, Shreya Buddharaju, Damien Kula, Soumyadeep Das, Hanyang Frank Liu, Faye Fei Mo, Rui Zhang, Wenpeng Yin. 2026. Bridging the Know-Act Gap via Task-Level Autoregressive Reasoning. arXiv preprint arXiv:2603.22619.

Bowen Jiang, Taiwei Shi, **Ryo Kamoi**, Yuan Yuan, Camillo J Taylor, Longqi Yang, Pei Zhou, Sihao Chen. 2026. One Model, All Roles: Multi-Turn, Multi-Agent Self-Play Reinforcement Learning for Conversational Social Intelligence. arXiv preprint arXiv:2602.03109.

Yusen Zhang, Wenliang Zheng, Aashrith Madasu, Peng Shi, **Ryo Kamoi**, Hao Zhou, Zhuoyang Zou, Shu Zhao, Sarkar Snigdha Sarathi Das, Vipul Gupta, Xiaoxin Lu, Nan Zhang, Ranran Haoran Zhang, Avitej Iyer, Renze Lou, Wenpeng Yin, Rui Zhang. 2025. HRScene: How Far Are VLMs from Effective High-Resolution Image Understanding? *ICCV 2025*.

Renze Lou, Hanzi Xu, Sijia Wang, Jiangshu Du, **Ryo Kamoi**, Xiaoxin Lu, Jian Xie, Yuxuan Sun, Yusen Zhang, Jihyun Janice Ahn, Hongchao Fang, Zhuoyang Zou, Wenchao Ma, Xi Li, Kai Zhang, Congying Xia, Lifu Huang, Wenpeng Yin. 2025. AAAR-1.0: Assessing AI's Potential to Assist Research. *ICML, Best Paper Award at the 2nd AI4Research workshop @ AAAI 2025.*

Sarkar Snigdha Sarathi Das, **Ryo Kamoi**, Bo Pang, Yusen Zhang, Caiming Xiong, Rui Zhang. 2025. GReaTer: Gradients over Reasoning Makes Smaller Language Models Strong Prompt Optimizers. *ICLR 2025*.

Jihyun Janice Ahn, **Ryo Kamoi**, Lu Cheng, Rui Zhang, Wenpeng Yin. 2024. Direct-Inverse Prompting: Analyzing LLMs' Discriminative Capacity in Self-Improving Generation. *arXiv preprint arXiv:2407.11017*.

Yilun Zhao, Yitao Long, Hongjun Liu, Linyong Nan, Lyuhao Chen, **Ryo Kamoi**, Yixin Liu, Xiangru Tang, Rui Zhang, Arman Cohan. 2023. DocMath-Eval: Evaluating Numerical Reasoning Capabilities of LLMs in Understanding Long Documents with Tabular Data. *ACL 2024.*

Yusen Zhang, Nan Zhang, Yixin Liu, Alexander Fabbri, Junru Liu, **Ryo Kamoi**, Xiaoxin Lu, Caiming Xiong, Jieyu Zhao, Dragomir Radev, Kathleen McKeown, Rui Zhang. 2024. Fair Abstractive Summarization of Diverse Perspectives. *NAACL 2024.*